STRATIFICATION BY FINITE ETALE-COVERED SCHEMES OVER QCQS SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 3964-3999, following the statement at 3949-3962.

Editorial note. The single 'some details omitted' covers the base-change pullback of the stratification (closed immersions, fp, finite etale, thickenings are stable under base change); flagged in the notes.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 3973 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Base change of algebraic spaces. Spaces, Definition spaces-definition-base-change and Properties of Spaces, Definition spaces-properties-definition-separated (cross-chapter, cited at ~3965-3967). Views X as an algebraic space over Spec(Z).
- Approximation by finite presentation spaces. proposition-approximate (same chapter, published as problem 1743 tag 07SU, cited at ~3968-3970). Affine X -> X_0 with X_0 fp over Z.
- A finite etale-covered scheme over a dense open. lemma-there-is-a-scheme-finite-over-refined (same chapter, cited at ~3983-3985). Finite surjective Y -> X, finite etale over a dense open U.

Source reference. Original lemma 0GUN: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 3911–3911. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original proposition 07SU: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 2328–2328. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 03I5: focused-seeds/geometry/S9-spaces-limits-there-is-a-scheme-finite-over-filtered/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original reference 03BS: focused-seeds/geometry/S9-spaces-limits-there-is-a-scheme-finite-over-filtered/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be a quasi-compact and quasi-separated algebraic space over S. There exists a $t \geq 0$ and closed subspaces*

$$X \supset Z_0 \supset Z_1 \supset \ldots \supset Z_t = \emptyset$$

such that $Z_i \to X$ is of finite presentation, $Z_0 \subset X$ is a thickening, and for each $i = 0, \ldots t-1$ there exists a scheme Y_i, a surjective, finite, and finitely presented morphism $Y_i \to Z_i$ which is finite étale over $Z_i \setminus Z_{i+1}$.

Proof. We may view X as an algebraic space over $\mathrm{Spec}(\mathbf{Z})$, see Spaces, Definition 03I5 and Properties of Spaces, Definition 03BS. Thus we may apply Proposition 07SU. It follows that we can find an affine morphism $X \to X_0$ with X_0 of finite presentation over $\mathbf{Z}$. If we can prove the lemma for X_0, then we can pull back the stratification and the morphisms to X and get the result for X; some details omitted. This reduces us to the case discussed in the next paragraph.

Assume X is of finite presentation over $\mathbf{Z}$. Then X is Noetherian and $|X|$ is a Noetherian topological space (with finitely many irreducible components) of finite dimension. Hence we may use induction on $\dim(|X|)$. Any finite morphism towards X is of finite presentation, so we can ignore that requirement in the rest of the proof. By Lemma 0GUN there exists a surjective finite morphism $Y \to X$ which is finite étale over a dense open $U \subset X$. Set $Z_0 = X$ and let $Z_1 \subset X$ be the reduced closed subspace with $|Z_1| = |X| \setminus |U|$. By induction we find an integer $t \geq 0$ and a filtration

$$Z_1 \supset Z_{1,0} \supset Z_{1,1} \supset \ldots \supset Z_{1,t} = \emptyset$$

by closed subspaces, where $Z_{1,0} \to Z_1$ is a thickening and there exist finite surjective morphisms $Y_{1,i} \to Z_{1,i}$ which are finite étale over $Z_{1,i} \setminus Z_{1,i+1}$. Since Z_1 is reduced, we have $Z_1 = Z_{1,0}$. Hence we can set $Z_i = Z_{1,i-1}$ and $Y_i = Y_{1,i-1}$ for $i \geq 1$ and the lemma is proved. □